
Discrete Phenotype Tokens Bridge Clinical Notes and EHR Sequences for Trustworthy Mortality Prediction

Zeyu Gan

University of North Carolina at Chapel Hill
ganzeyu@unc.edu

Abstract

Structured EHR sequences capture longitudinal diagnostic trajectories, yet many prognostic signals relevant to high-acuity outcomes are documented only in free-text discharge summaries. A natural response is to fuse dense note embeddings with structured representations; however, in our small-parameter EHR model setting, a continuous projection approach (the NoteProjector) suffered from representation collapse and failed to improve broad ICD prediction. We present Delphi-Px, a discrete alternative that extracts 33 clinical phenotype tokens from discharge summaries and inserts them directly into the event timeline. These tokens, which cover cancer status, psychiatric crisis, substance-use episodes, and critical-care states, are intentionally narrow in scope: they target mortality-relevant clinical states that are poorly captured by structured ICD codes alone. On MIMIC-IV, Delphi-Px raises the Death AUC from 0.680 (achieved by the structured-only Delphi-Base) to 0.721, corresponding to an absolute gain of +0.041, whereas static logistic regression and histogram gradient boosting baselines reach only 0.558 and 0.561, respectively. SHAP analysis confirms that all 33 tokens carry positive mean attribution for Death, with the palliative-care, DNR, ICU, intubation, sepsis, and acute kidney injury tokens supplying clinically interpretable risk evidence. This gain is accompanied by a modest decline in broad ICD chapter prediction (overall AUC from 0.849 to 0.841), which points to task-specific specialization rather than uniform improvement.

1 Introduction

Clinical prediction increasingly demands the integration of heterogeneous data sources (Kline et al., 2022). Structured EHR sequences record longitudinal diagnoses, procedures, laboratory values, and demographics, while narrative discharge summaries capture clinical judgments, goals-of-care decisions, and acute events that are often most relevant to prognosis (Garriga et al., 2023; Rajkomar et al., 2018). EHR sequence models such as BEHRT, Med-BERT, and Delphi have demonstrated that structured event histories alone can support future disease prediction (Li et al., 2020; Rasmy et al., 2021; Shmatko et al., 2025). ICD-coded histories are well suited to chronic disease trajectories (Pedersen et al., 2023; Siggaard et al., 2020); however, mortality risk frequently depends on signals such as comfort-measures-only designations, do-not-resuscitate orders, intubation status, delirium, or active sepsis, all of which may appear in discharge summaries without timely structured coding (Kumar et al., 2023; Cooley-Rieders and Zheng, 2021).

A common approach is to encode notes with a clinical language model and to fuse the resulting dense vectors with structured EHR representations (Lyu et al., 2022; Liu et al., 2022; Bardak and Tan, 2021). ClinicalBERT and Clinical-Longformer adapt transformer encoders to clinical text of varying length (Huang et al., 2019; Li et al., 2022), thereby extending general transformer language modeling (Devlin et al., 2019). We initially pursued this strategy through a NoteProjector module

that projected 768-dimensional Clinical-Longformer embeddings into the 192-dimensional Delphi token space. The NoteProjector was computationally feasible but proved difficult to align with the discrete token geometry: the projected representations exhibited near-collapse (with pairwise cosine similarity of 0.9977), required diversity regularization to partially recover variance, and produced no net gain in broad ICD prediction. Equally important, the continuous representation offered no token-level audit trail for clinical review.

We therefore investigate a simpler hypothesis: if much of the note-derived mortality signal resides in a small set of high-acuity clinical states, then those states can be extracted explicitly and treated as ordinary events in the sequence (De Silva et al., 2021; Lyu et al., 2022; Yang et al., 2021; Zhang et al., 2022; Bardak and Tan, 2021). The resulting model, Delphi-Px, extends Delphi-Base with 33 discrete clinical phenotype tokens that are extracted from discharge summaries by a rule-based pipeline with section-awareness and negation handling. The tokens are inserted at discharge time, the embedding matrix is expanded, and the model is fine-tuned with the same autoregressive next-event objective and Weibull time-to-event regularization that are used by the base model.

This paper makes three empirical contributions. First, it presents a discrete text-to-sequence fusion mechanism that preserves the causal event interface of Delphi while adding auditable note-derived states. Second, it describes the leakage controls, including section filtering, negation handling, temporal intent guards, and the exclusion of phenotype tokens from future test labels, that are necessary for prospective mortality modeling. Third, it quantifies a task-specific fine-tuning trade-off in which the Death AUC improves by +0.041 while the average broad ICD chapter AUC declines by approximately 0.010. All results are limited to the MIMIC-IV held-out test split ($N = 9,915$); external validation on an independent hospital system is required before clinical deployment.

2 Related work

EHR sequence modeling. Transformer-based approaches to structured EHR data, including BEHRT, Med-BERT, and Delphi, represent diagnoses and clinical events as discrete codes and optimize future-event prediction (Li et al., 2020; Rasmy et al., 2021; Shmatko et al., 2025). Delphi combines generative event modeling with continuous age encodings and a time-to-event prediction head (Shmatko et al., 2025).

Multimodal clinical AI. Multimodal clinical systems combine structured records with notes, imaging, waveforms, or laboratory data through early fusion, late fusion, or dense representation alignment (Acosta et al., 2022; Soenksen et al., 2022). Clinical note encoders such as ClinicalBERT and Clinical-Longformer produce dense representations of unstructured clinical text (Huang et al., 2019; Li et al., 2022). These methods can be effective but often require separate encoders and may obscure the clinical evidence behind individual predictions. The NoteProjector described in Section 1 serves as an internal point of comparison: it attempted dense note-vector alignment but did not produce interpretable token-level evidence.

Interpretable survival and risk modeling. Our model uses a Weibull time-to-event head as a statistical regularizer (Weibull, 1951) and evaluates the resulting risk predictions with binary AUC, C-index, 30-day mortality AUC, and SHAP attribution. The estimation of AUC follows the method of DeLong (DeLong et al., 1988), the survival ranking follows the concordance framework of Harrell (Harrell et al., 1996), and the feature attribution uses SHAP (Lundberg and Lee, 2017).

3 Methodology

3.1 Architecture of Delphi-Base

The structured baseline, Delphi-Base, builds on the Delphi backbone (Shmatko et al., 2025) with 5.83M parameters and 1,537 model-space tokens. It uses a GPT-2-style causal Transformer with 12 layers, 12 attention heads, an embedding dimension of 192, a block size of 128, and four Global Memory summary positions. Stored event IDs are shifted by +1 in model space. The vocabulary comprises demographic tokens, structured ICD-10 diagnosis codes, laboratory tokens, a no-event token, and a Death token. The cross-entropy loss excludes padding, Global Memory placeholders,

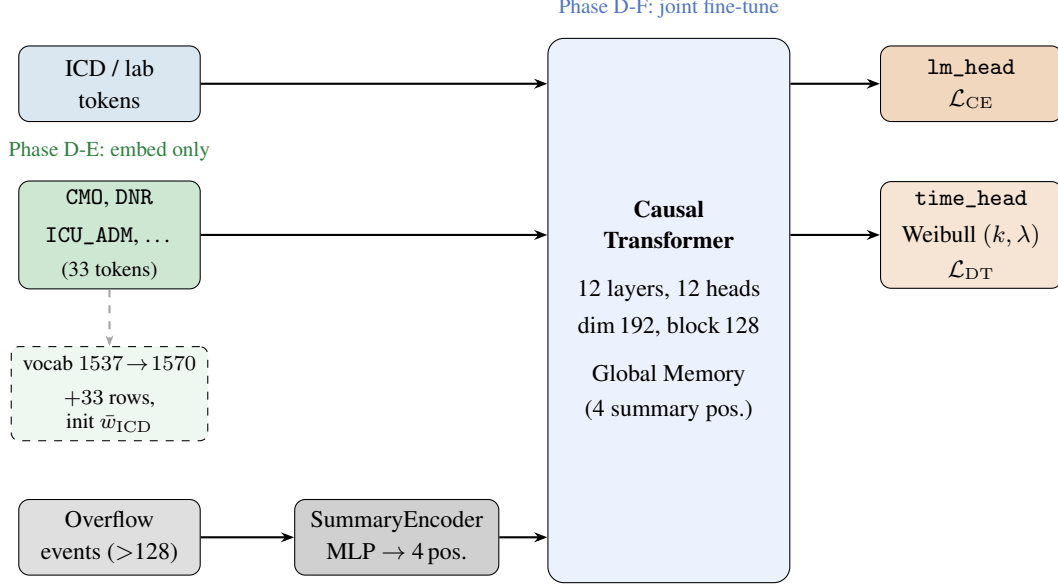


Figure 1: Architecture of Delphi-Px. Three input streams enter the causal Transformer: structured ICD/lab tokens (blue), 33 new phenotype tokens (green), and up to 64 overflow events compressed into 4 Global Memory positions by the SummaryEncoder MLP. The backbone feeds two heads: `lm_head` ($\mathcal{L}_{CE}$, next-event prediction) and `time_head` ($\mathcal{L}_{DT}$, Weibull survival). The dashed arrow indicates that the vocabulary is expanded from 1,537 to 1,570 by appending 33 embedding rows initialized near $\bar{w}_{ICD}$. Phase D-E (1,000 steps) trains only those rows; Phase D-F (3,000 steps) fine-tunes all parameters.

sex, BMI, and laboratory tokens, thereby restricting the autoregressive objective to clinical event prediction.

The Global Memory mechanism compresses up to 64 overflow events lying outside the current 128-token window into four summary embeddings through a SummaryEncoder MLP, thus preserving long-history information without enlarging the attention window. Figure 1 shows the full architecture and the two-phase fine-tuning protocol.

3.2 Statistical time-to-event modeling

In addition to the next-event cross-entropy, Delphi-Base includes a Weibull time-to-event head (Weibull, 1951). For an inter-event interval $t > 0$, shape k , and scale λ , the Weibull density is given by

$$f(t; k, \lambda) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} \exp \left[- \left(\frac{t}{\lambda} \right)^k \right]. \quad (1)$$

The shape parameter permits non-constant hazard rates, and the exponential distribution is recovered when $k = 1$. The model predicts the raw parameters r_k and r_λ and applies an exponential reparameterization with clamping:

$$k = \exp(\text{clip}(r_k, -6, 4)), \quad \log(t/\lambda) = \log t - r_\lambda. \quad (2)$$

The log-likelihood is evaluated entirely in log space, and each per-token contribution is clamped to a minimum of -50 nats before averaging:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CE} + \mathcal{L}_{DT}. \quad (3)$$

These numerical safeguards were introduced in the Delphi backbone to prevent float32 overflow in the time head.

3.3 Discrete clinical phenotyping

Delphi-Px adds 33 phenotype tokens to the Delphi-Base vocabulary across four clinical domains: cancer status and treatment (14 tokens), psychiatric crisis (8), substance use (4), and high-acuity critical care (7). Representative tokens include `CANCER_STAGE_IV`, `CANCER_METASTATIC`, `SUICIDAL_IDEATION_PRESENT`, `ALCOHOL_WITHDRAWAL_ACTIVE`, `SEPSIS_PRESENT`, `INTUBATED_DURING_STAY`, `COMFORT_MEASURES_ONLY`, `DNR_PRESENT`, and `ICU_ADMISSION`. The set is intentionally non-comprehensive: it targets mortality-relevant states that are systematically absent from, delayed in, or ambiguous within structured ICD histories.

Extraction proceeds by applying regular expressions with clinical-context guards to discharge summaries. Notes are segmented into named sections (history of present illness, past medical history, hospital course, assessment and plan, discharge diagnoses, among others), and follow-up and discharge-medication sections are excluded in order to avoid capturing future plans. An 80-character lookback window placed before each match suppresses negated mentions (*e.g.*, “no suicidal ideation”), and a dedicated token is reserved for explicitly denied ideation. Treatment-received tokens are suppressed when the lookback context suggests a future plan, while treatment-planned tokens follow the inverse logic. Cancer-staging tokens require a cancer keyword within 200 characters. Only the first match per token per note is retained.

Across 331,793 discharge summaries, the extractor identifies at least one phenotype token in 136,865 notes (41.3%), yielding 242,601 token occurrences. The five most frequent tokens are `ICU_ADMISSION` (29,405), `AKI_PRESENT` (23,966), `DELIRIUM_PRESENT` (22,788), `CANCER_RESECTED` (20,082), and `INTUBATED_DURING_STAY` (19,255). The processing was performed on 30 workers and completed in 188 seconds.

3.4 Leakage prevention

The phenotype tokens are intended to serve as prospective prognostic context rather than as proxies for future outcomes. The pipeline applies three safeguards. Section filtering excludes discharge instructions and medication sections, in which references to future plans are common. Negation handling suppresses clinically opposite statements such as denied suicidal ideation. Temporal and intent guards distinguish treatment that has already been received from treatment that is merely planned. During dataset construction, phenotype tokens are inserted into `train.bin`, `val.bin`, and `test_input.bin` but not into `test_future.bin`, so that the tokens serve as input context for test patients without appearing as prediction labels. These three controls (section filtering via `AVOID_SECTIONS = {followup, discharge_meds, discharge_instr}`, negation handling, and temporal intent guards) constitute the complete leakage-prevention mechanism; no additional phrase-level exclusion rules for post-mortem language are applied.

3.5 Two-phase fine-tuning

The Delphi-Base checkpoint contains a word-token embedding matrix of shape (1537, 192). Delphi-Px expands this to (1570, 192) by appending 33 rows initialized at the mean ICD embedding plus Gaussian noise ($\sigma = 0.01$). The tied output projection is expanded in the same manner.

Training proceeds in two phases. Phase D-E is a 1,000-step embedding warmup with the backbone frozen: only the 33 new phenotype rows in `wte.weight` and `lm_head.weight` receive gradient updates, and the old rows are masked after unscaling, which yields an effective parameter count of $33 \times 192 = 6,336$. This phase uses AdamW at a learning rate of 10^{-3} . Phase D-F is a 3,000-step joint fine-tune that starts from the D-E checkpoint, with all 5,837,570 parameters trainable at a learning rate of 10^{-5} . Both phases use a batch size of 128, a block size of 128, a weight decay of 0.1, gradient clipping of 1.0, and a single NVIDIA L40S GPU (46 GB).

4 Experimental setup

Dataset and splits. The experiments use MIMIC-IV-derived structured EHR sequences along with 331,793 discharge summaries (Johnson et al., 2023). Table 1 summarizes the cohort. Patients are included if they have at least two admissions spanning more than 30 days, or a single admission with a length of stay exceeding 7 days; in addition, each patient must have at least three distinct

disease-code events after tokenization. No explicit age filter is applied beyond what is already present in the source MIMIC-IV tables. ICD-9 and ICD-10 codes are distinguished by the `icd_version` field rather than by calendar date, because the admission dates in MIMIC-IV are temporally shifted. Patients are split 70/20/10 (train/validation/test) by patient identifier with a random seed of 42. For the survival evaluation, the test-set cutoff age is defined as the last admission age minus 180 days.

Table 1: Cohort definition and data split statistics.

Criterion / statistic	Value
Admission filter	≥ 2 admissions, span > 30 d; <i>or</i> single admission, LOS > 7 d
Minimum disease-code events	3
ICD versioning	<code>icd_version</code> field (9 / 10)
Age filter	None (MIMIC-IV dates are temporally shifted)
Split ratio (train / val / test)	70% / 20% / 10%
Random seed	42
Training patients	69,407
Validation patients	19,830
Test patients	9,915
Train events	2,733,096
Validation events	782,869
Test-input events	221,959
Test-future events	173,764
Discharge summaries (total)	331,793

Baselines. We compare Delphi-Px against three baselines: logistic regression on age, sex, and the 33 phenotype indicators; histogram gradient boosting (HGBT) on the same features together with an ICD bag-of-words; and Delphi-Base Phase 2b, which is the structured-only sequence model. The HGBT baseline uses the `HistGradientBoostingClassifier` of scikit-learn (Pedregosa et al., 2011), which handles class imbalance natively through `class_weight='balanced'`.

Evaluation. The primary endpoint is future Death prediction, which is reported as AUC. Chapter-level AUC values are computed with the corrected `evaluate_auc.py` protocol using the DeLong method (DeLong et al., 1988), the same post-2015 test split, and the same filter threshold for Delphi-Base and Delphi-Px. We also report the C-index (Harrell et al., 1996) and the 30-day mortality AUC derived from the Weibull time head. Explainability is assessed using SHAP Partition (Lundberg and Lee, 2017) on 500 validation patients with at least one phenotype token in the last 64-event window, targeting the Death logit at the final sequence position.

5 Results

5.1 Interpretable mortality prediction

Table 2 reveals a clear staircase pattern. Static phenotype features and bag-of-words features fall well short of the structured sequence baseline: HGBT improves over logistic regression by only +0.003. Delphi-Base raises the Death AUC to 0.680, which confirms that temporal sequence modeling captures information that is absent from static event counts. The addition of discrete phenotype tokens further raises the Death AUC to 0.721, an absolute gain of +0.041 over the structured-only model.

5.2 Task shift and the Pareto frontier

The joint fine-tuning specializes the model toward mortality risk. As Table 3 and Figure 2 show, the Death AUC increases by +0.041 while the AUC for the 17 ICD chapters declines by an average of 0.010, with the largest drops being observed for Pregnancy and Childbirth (−0.016), Musculoskeletal Diseases (−0.015), and Skin Diseases (−0.014). The frozen-backbone D-E checkpoint shows a negligible ICD decline but only a +0.004 Death gain, which localizes the improvement to backbone

Table 2: Death prediction performance. The static baselines use no sequence ordering; Delphi-Base uses only the structured sequence; Delphi-Px adds discrete phenotype tokens.

Model	Death AUC	Input representation
Logistic regression	0.558	Age, sex, 33 phenotype indicators
HGBT	0.561	Age, sex, phenotype indicators, ICD bag-of-words
Delphi-Base	0.680	Structured EHR sequence
Delphi-Px	0.721	Structured sequence plus phenotype tokens

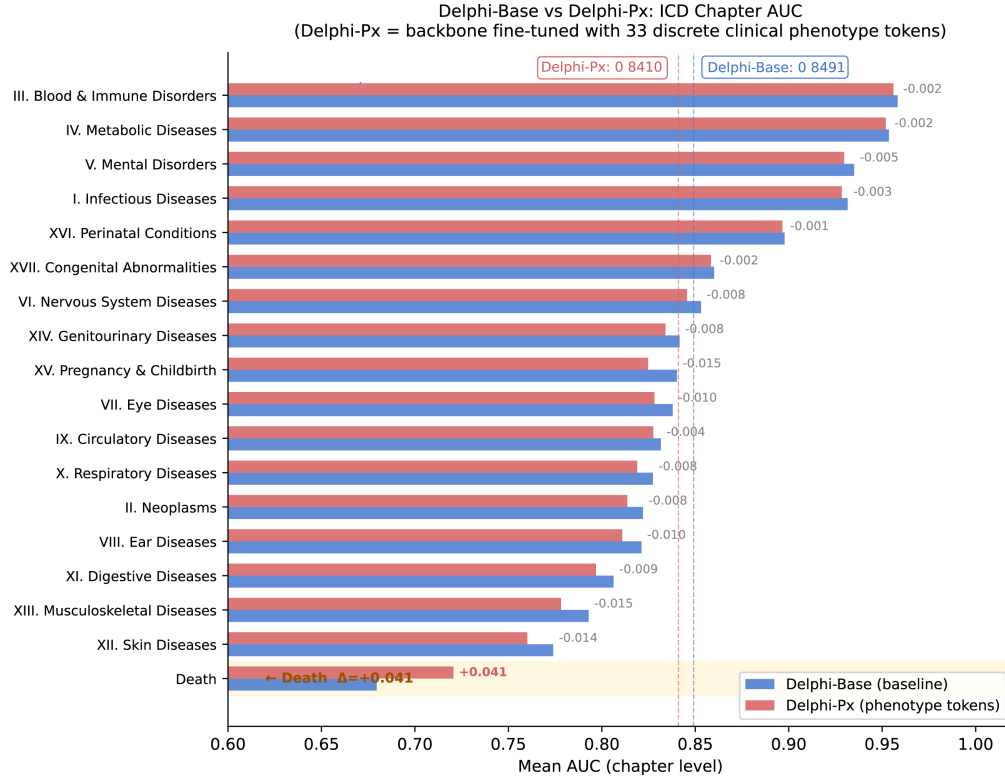


Figure 2: Chapter-level AUC comparison between Delphi-Base and Delphi-Px. The largest gain is observed for Death prediction, while the broad ICD chapters show modest declines after joint fine-tuning.

plasticity: the model must adapt its attention routing in order to use the phenotype tokens, and this adaptation slightly erodes broad ICD prediction. We therefore view Delphi-Px as a task-specialized checkpoint rather than as a uniformly superior replacement for Delphi-Base.

5.3 Trustworthy explainability via SHAP

The SHAP analysis identifies a clinically coherent mechanism behind the Death AUC gain. All 33 phenotype tokens have positive mean SHAP values for Death prediction, with an average of +0.189 across the tokens. The phenotype tokens with the highest attribution are COMFORT_MEASURES_ONLY (+0.408), ICU_ADMISSION (+0.330), INTUBATED_DURING_STAY (+0.328), ALCOHOL_WITHDRAWAL_ACTIVE (+0.307), and DNR_PRESENT (+0.256). Among the top 30 tokens by absolute SHAP value for Death, five are phenotype tokens, even though phenotype tokens make up only 33 of 1,570 model-space entries. This concentration supports the claim that text-derived states carry disproportionately high mortality-relevant information.

The final checkpoint also provides embedding-space evidence that the phenotype tokens are placed coherently. Their mean L2 norm is 0.914, compared with 1.415 ± 0.315 for the ICD tokens, which is consistent with high-specificity probes occupying a lower-magnitude region of the embedding

Table 3: Overall and Death AUC across the backbone-frozen and the joint-fine-tuned Delphi-Px checkpoints.

Model	Overall AUC	Death AUC	Δ overall	Δ Death
Delphi-Base Phase 2b	0.8491	0.6796	–	–
Delphi-Px D-E, frozen backbone	0.8468	0.6840	−0.0023	+0.0044
Delphi-Px D-F, joint fine-tune	0.8410	0.7206	−0.0081	+0.0411

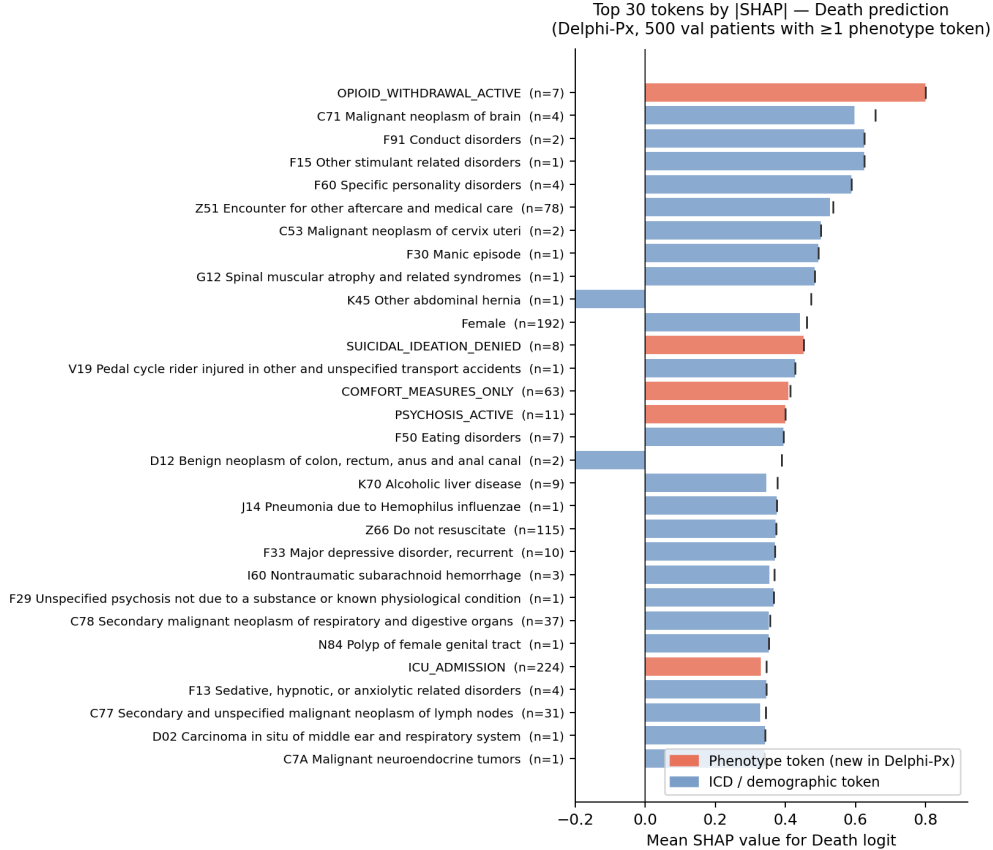


Figure 3: The top tokens by absolute SHAP value for Death prediction in Delphi-Px. The new phenotype tokens are over-represented among the high-attribution tokens relative to their share of the vocabulary.

space. The `CANCER_STAGE_I` token, whose cross-entropy loss was inadvertently suppressed by an `ignore_tokens` bug (Limitation 4), has a norm of 0.855, which lies within the phenotype cluster. The gradient signal from the input-side attention during Phase D-F was apparently sufficient for partial convergence even without a prediction target.

5.4 Ablation and survival statistics

The Weibull head shows a modest improvement but remains substantially weaker than the binary Death logit. Delphi-Px raises the C-index from 0.494 to 0.533 and the 30-day mortality AUC from 0.486 to 0.525. These results should not be read as evidence of calibrated survival modeling: the binary Death AUC of 0.721 far exceeds the time-head metrics, and the near-chance 30-day AUC of Delphi-Base (0.486) reflects a degenerate high shape parameter ($\hat{k} \approx 31.6$) that compresses the Weibull CDF toward zero for all patients. The Weibull head in this architecture is best understood as a time-to-event regularizer; extending it to a proper survival model would require censoring-aware objectives such as partial-likelihood-based losses (Cox, 1972).

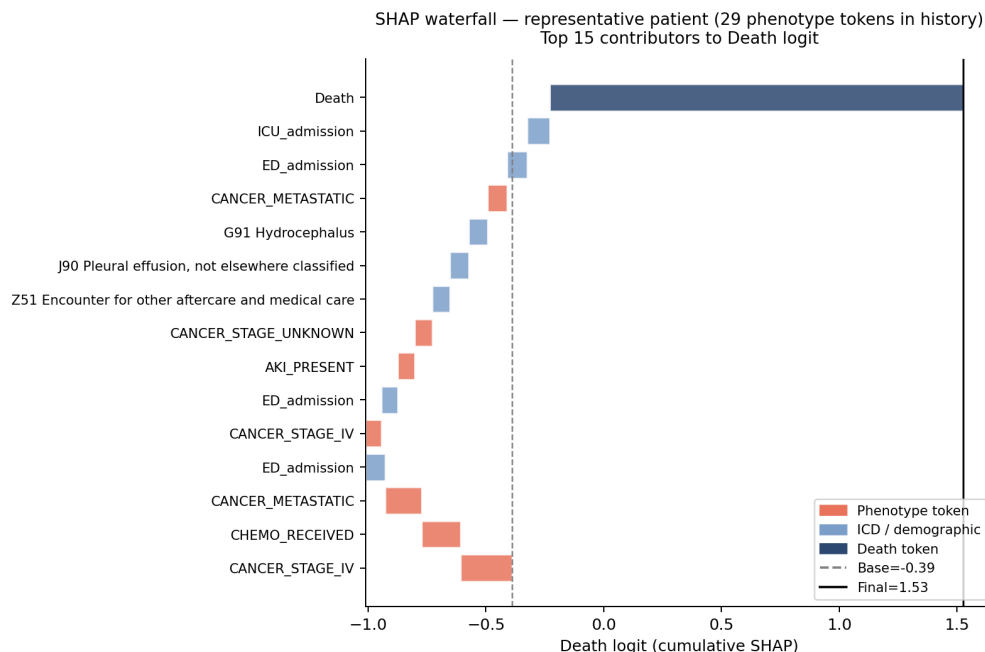


Figure 4: A representative patient-level SHAP waterfall for Death prediction. The attribution trace provides an auditable token-level explanation.

Table 4: Survival-head metrics on the held-out test set. The confidence intervals are obtained from 1,000-resample stratified bootstrap, as reported in the technical report.

Metric	Delphi-Base	Delphi-Px	Δ
Binary Death AUC	0.680	0.721	+0.041
C-index	0.494 [0.478, 0.512]	0.533 [0.513, 0.551]	+0.039
30-day mortality AUC	0.486 [0.464, 0.509]	0.525 [0.501, 0.548]	+0.039

The discrete-token design also compares favorably with the continuous NoteProjector in terms of interpretability. In preliminary runs, the NoteProjector achieved a Death AUC gain of roughly +0.04–0.05, which is comparable with Delphi-Px, but it required diversity regularization to counter representation collapse and provided no token-level attribution. Since no single stable NoteProjector checkpoint was retained, it is not used as a formal baseline; nonetheless, the qualitative contrast (a similar predictive gain coupled with substantially less interpretability) motivates the discrete-token approach.

6 Discussion

Discrete extraction versus continuous alignment. Delphi-Px suggests that, for small EHR sequence models, clinically curated discrete extraction can serve as a more stable bridge between notes and structured timelines than dense embedding alignment. The method gives up broad recall over full note semantics but gains transparent token identities, direct compatibility with the autoregressive vocabulary, and straightforward SHAP-based auditability.

Clinical deployment. The results support maintaining two checkpoints for different purposes. Delphi-Base Phase 2b remains the stronger broad-spectrum ICD surveillance model (overall AUC 0.8491). Delphi-Px is better suited to mortality risk stratification, palliative-care decision support research, or high-acuity cohorts in which named note-derived states are central to risk. A single checkpoint should not be expected to optimize both broad disease surveillance and mortality specialization simultaneously.

Generalization and safety. All results come from MIMIC-IV. The extractor is calibrated to the discharge-summary language in that dataset and may need to be adapted for other hospitals, outpatient notes, non-English text, or different documentation practices. Any deployment would require external validation, prospective calibration, subgroup analysis, and clinician-facing review of the extracted evidence. Subgroup and fairness evaluation across sex, age, race, and insurance status is absent from this study and is a prerequisite for clinical use.

7 Limitations

The 33-token dictionary is intentionally narrow and does not cover cardiovascular interventions, renal replacement therapy, ventilator weaning, chronic disease staging, or outpatient phenotypes. Rule-based extraction is oriented toward precision and will miss atypical phrasings. Phenotype timestamps are assigned at discharge time rather than at the exact within-admission event time, which introduces imprecision for long admissions. The metadata of Delphi-Base inadvertently masked `CANCER_STAGE_I` in the cross-entropy loss; this token is rare (with 194 occurrences, 0.08% of phenotype events) and its embedding norm falls within the phenotype cluster, but the issue should be corrected before retraining. SHAP was computed only for Death and only on validation patients with at least one phenotype token in the last 64-event window. There is no external validation beyond the MIMIC-IV held-out split. No model card (Mitchell et al., 2019) has been prepared; responsible deployment would require documentation of the intended use, out-of-scope uses, the MIMIC-IV data access requirements under the PhysioNet Data Use Agreement, and clinical validation. The system is trained on a single US academic medical center dataset, and its performance on community hospitals, non-English documentation, or other coding practices is unknown.

8 Conclusion

Delphi-Px shows that discrete clinical phenotype tokens extracted from discharge summaries can serve as a stable and interpretable bridge between clinical notes and EHR sequence models. On MIMIC-IV, the proposed approach improves the Death AUC by +0.041 over the structured-only Delphi-Base baseline, and SHAP attribution traces the gain to clinically meaningful high-acuity states rather than to opaque dense note features. The improvement is task-specific, trading a modest amount of broad ICD prediction for a useful mortality specialization. This makes Delphi-Px a promising proof of concept for trustworthy text-to-token fusion, with external validation, broader phenotype coverage, and calibrated survival modeling as natural next steps.

References

- Julian N. Acosta, Guido J. Falcone, Pranav Rajpurkar, and Eric J. Topol. Multimodal biomedical AI. *Nature Medicine*, 28:1773–1784, 2022. DOI: 10.1038/s41591-022-01981-2.
- D. R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2):187–202, 1972. DOI: 10.1111/j.2517-6161.1972.tb00899.x.
- Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3):837–845, 1988. DOI: 10.2307/2531595.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019. DOI: 10.18653/v1/N19-1423.
- Frank E. Harrell Jr., Kerry L. Lee, and Daniel B. Mark. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Statistics in Medicine*, 15(4):361–387, 1996. DOI: 10.1002/(SICI)1097-0258(19960229)15:4<361::AID-SIM168>3.0.CO;2-4.
- Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. ClinicalBERT: Modeling clinical notes and predicting hospital readmission. *arXiv preprint arXiv:1904.05342*, 2019.

- Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-wei H. Lehman, Leo A. Celi, and Roger G. Mark. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023. DOI: 10.1038/s41597-022-01899-x.
- Yikuan Li, Shishir Rao, Jose Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. BEHRT: Transformer for electronic health records. *Scientific Reports*, 10:7155, 2020. DOI: 10.1038/s41598-020-62922-y.
- Yikuan Li, Ramsey M. Wehbe, Faraz S. Ahmad, Hanyin Wang, and Yuan Luo. Clinical-Longformer and Clinical-BigBird: Transformers for long clinical sequences. *arXiv preprint arXiv:2201.11838*, 2022. DOI: 10.48550/arXiv.2201.11838.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine*, 4:86, 2021. DOI: 10.1038/s41746-021-00455-y.
- Artem Shmatko, Alexander Wolfgang Jung, Kumar Gaurav, Soren Brunak, Laust Hvas Mortensen, Ewan Birney, Tom Fitzgerald, and Moritz Gerstung. Learning the natural history of human disease with generative transformers. *Nature*, 647:248–256, 2025. DOI: 10.1038/s41586-025-09529-3.
- Luis R. Soenksen, Yu Ma, Cynthia Zeng, Leonard Boussieux, Kimberly Villalobos Carballo, Liangyuan Na, Holly M. Wiberg, Michael L. Li, Ignacio Fuentes, and Dimitris Bertsimas. Integrated multimodal artificial intelligence framework for healthcare applications. *npj Digital Medicine*, 5:149, 2022. DOI: 10.1038/s41746-022-00689-4.
- Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FACCT)*, pages 220–229, 2019. DOI: 10.1145/3287560.3287596.
- Waloddi Weibull. A statistical distribution function of wide applicability. *Journal of Applied Mechanics*, 18(3):293–297, 1951. DOI: 10.1115/1.4010337.
- Batuhan Bardak and Mehmet Tan. Improving clinical outcome predictions using convolution over medical entities with multimodal learning. *Artificial Intelligence in Medicine*, 117:102112, 2021. DOI: 10.1016/j.artmed.2021.102112.
- Kristin Cooley-Rieders and Kai Zheng. Physician documentation matters: Using natural language processing to predict mortality in sepsis. *Intelligence-Based Medicine*, 5:100028, 2021. DOI: 10.1016/j.ibmed.2021.100028.
- Kathryn De Silva, Natalie Mathews, Helena Teede, Andrew Forbes, Dorota Jönsson, Ryan Demmer, and Joanne Enticott. Clinical notes as prognostic markers of mortality associated with diabetes mellitus following critical care: A retrospective cohort analysis using machine learning and unstructured big data. *Computers in Biology and Medicine*, 132:104305, 2021. DOI: 10.1016/j.combiomed.2021.104305.
- Rita Garriga, Tomas Buda, Joao Guerreiro, Javier Iglesias, Inigo Aguerri, and Aleksandar Matić. Combining clinical notes with structured electronic health records enhances the prediction of mental health crises. *Cell Reports Medicine*, 4:101260, 2023. DOI: 10.1016/j.xcrm.2023.101260.
- Adrienne Kline, Hanyin Wang, Yikuan Li, Saya Dennis, Margaret Hutch, Zhenxing Xu, Fei Wang, Feixiong Cheng, and Yuan Luo. Multimodal machine learning in precision health: A scoping review. *npj Digital Medicine*, 5:171, 2022. DOI: 10.1038/s41746-022-00712-8.

- Anil Kumar, Nicholas Hammond, Steven Grattan, Simon Finfer, and Anthony Delaney. Accuracy of International Classification of Disease coding methods to estimate sepsis epidemiology: A scoping review. *Journal of Intensive Care Medicine*, 39(1):3–11, 2023. DOI: 10.1177/08850666231192371.
- Shengpu Liu, Xiaoqian Wang, Yongkang Hou, Guanghui Li, Haoyue Wang, Hua Xu, Yang Xiang, and Buzhou Tang. Multimodal data matters: Language model pre-training over structured and unstructured electronic health records. *IEEE Journal of Biomedical and Health Informatics*, 27(2):504–514, 2022. DOI: 10.1109/JBHI.2022.3217810.
- Weijian Lyu, Xiaoyu Dong, Rachel Wong, Shijing Zheng, Karin Abell-Hart, Fei Wang, and Changhe Chen. A multimodal transformer: Fusing clinical notes with structured EHR data for interpretable in-hospital mortality prediction. In *AMIA Annual Symposium Proceedings*, pages 719–728, 2022. DOI: 10.48550/arXiv.2208.10240.
- Marlene G. Pedersen, Rune Eriksson, Rebecka Reguant, Cecilia Collin, Henrik L. Pedersen, Frederik H. Sørup, Cresta Simon, Abigail Birch, Malene R. Larsen, Alexander V. Nielsen, Katharina Belling, and Søren Brunak. A unidirectional mapping of ICD-8 to ICD-10 codes, for harmonized longitudinal analysis of diseases. *European Journal of Epidemiology*, 38:1043–1052, 2023. DOI: 10.1007/s10654-023-01027-y.
- Alvin Rajkomar, Eyal Oren, Kai Chen, Andrew M. Dai, Nissan Hajaj, Michaela Hardt, Peter J. Liu, Xiaobing Liu, Jake Marcus, Mimi Sun, Patrik Sundberg, Hector Yee, Kun Zhang, Yi Zhang, Gerardo Flores, Gavin E. Duggan, Jamie Irvine, Quoc Le, Kurt Litsch, Alexander Mossin, Justin Tansuwan, De Wang, James Wexler, Jimbo Wilson, Dana Ludwig, Samuel L. Volchenbourn, Katherine Chou, Michael Pearson, Srinivasan Madabushi, Nigam H. Shah, Atul J. Butte, Michael D. Howell, Claire Cui, Greg S. Corrado, and Jeff Dean. Scalable and accurate deep learning with electronic health records. *npj Digital Medicine*, 1:18, 2018. DOI: 10.1038/s41746-018-0029-1.
- Tue Siggaard, Rebecka Reguant, Ioanna G. Jørgensen, Agnete D. Haue, Mikkel Lademann, Alejandro Aguayo-Orozco, Jessica X. Hjaltelin, Anders B. Jensen, Karina Banasik, and Søren Brunak. Disease trajectory browser for exploring temporal, population-wide disease progression patterns in 7.2 million Danish patients. *Nature Communications*, 11:5024, 2020. DOI: 10.1038/s41467-020-18682-4.
- Hao Yang, Longbing Kuang, and Fei Xia. Multimodal temporal-clinical note network for mortality prediction. *Journal of Biomedical Semantics*, 12:3, 2021. DOI: 10.1186/s13326-021-00235-3.
- Jenny Zhang, Luis Trujillo, Arnav Tanwar, Julia Ive, Vibhor Gupta, and Yulan Guo. Clinical utility of automatic phenotype annotation in unstructured clinical notes: intensive care unit use. *BMJ Health & Care Informatics*, 29:e100519, 2022. DOI: 10.1136/bmjhci-2021-100519.

A Reproducibility details

Code. The full pipeline is publicly available at <https://github.com/AllyssaDrew/Delphi-Px-EHR-Foundation-model-with-MIMIC-IV>. The repository is organized into five stages. **(1) Structured preprocessing** (`preprocessing/02_preprocess_v5.py`, `preprocessing/04_make_mimic_labels.py`): converts raw MIMIC-IV tables into Delphi-Base binary event sequences. **(2) Continuous note-fusion experiment** (`multimodal_notes/`): builds Clinical-Longformer embeddings for discharge summaries and inserts a dense NOTE token into the sequences; this branch was ultimately abandoned due to representation collapse. **(3) Discrete phenotype token extraction** (`Phase_A/phenotype_dict.py`, `Phase_A/extract_phenotypes.py`, `Phase_B/run_extraction.py` under `clinical_phenotyping/`): applies 33 section-aware, negation-guarded regular expressions to all discharge summaries using 30 parallel workers (≈ 188 seconds on a 32-core CPU node). **(4) Dataset construction and fine-tuning** (`clinical_phenotyping/Phase_C/build_v61_dataset.py`, `clinical_phenotyping/Phase_D/expand_vocab_v5_to_v61.py`, `clinical_phenotyping/Phase_D/train_phase_d_e.sbatch`, `clinical_phenotyping/Phase_D/train_phase_d_f.sbatch`): merges phenotype events into the Delphi-Base sequences, expands the checkpoint vocabulary from 1,537 to 1,570 tokens, runs Phase D-E (1,000 steps, embedding warmup) followed by Phase D-F (3,000 steps,

joint fine-tune). **(5) Evaluation and explainability** (evaluation/analysis_wrapup.py, evaluation/bootstrap_ci.py, clinical_phenotyping/Phase_E/shap_v61.sbatch, clinical_phenotyping/Phase_E/plot_shap_v61.py, figures/05_figures.py): computes DeLong AUC, ML baselines, bootstrap confidence intervals, SHAP Partition attributions, and all manuscript figures. Production training configurations are in configs/final/.

Environment. Python 3.11, PyTorch 2.2.1 (CUDA 12.1), scikit-learn 1.4, SHAP 0.45, NumPy 1.26, Pandas 2.2, umap-learn 0.5.6. A complete requirements.txt and environment.yml (conda) are provided in the repository. Training used a single NVIDIA L40S GPU (46 GB VRAM). Phase D-E completes in <5 minutes, Phase D-F in <10 minutes, and the bootstrap CI (1,000 resamples) in <2 hours on CPU.

Random seeds. Patient split: random_state=42. Embedding initialization noise: torch.manual_seed(42). Bootstrap CI: numpy.random.default_rng(42).

Data access. MIMIC-IV and MIMIC-IV-Note are available via PhysioNet (<https://physionet.org/content/mimiciv/>, <https://physionet.org/content/mimic-iv-note/>) under the PhysioNet Credentialed Health Data License 1.5.0. Credentialed access requires completion of the CITI “Data or Specimens Only Research” training and a signed Data Use Agreement. No patient data is included in the repository. Instructions for obtaining access and configuring the required environment variables (MIMIC_DATA_DIR, MIMIC_NOTE_DIR, DELPHI_PROJECT_ROOT) are provided in docs/data_access.md.

B Conference checklist status

The checklist has been completed. All items have been answered; please refer to the checklist file for the corresponding justifications.